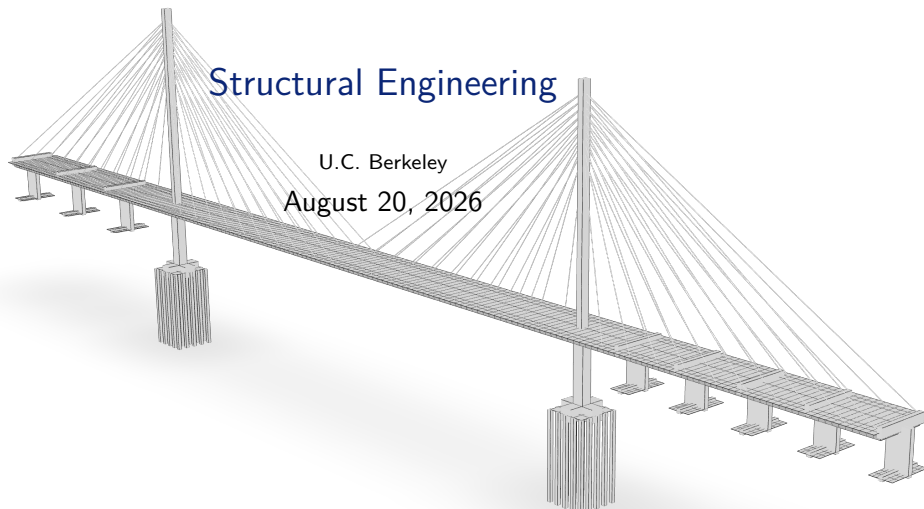




① Introduction

② Math

③ Modeling

④ Design

⑤ Research

Summary

Objectives

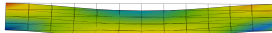
- Review some prerequisites for the structural engineering curriculum
- Provide some perspective

Comments

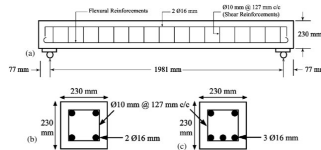
We'll do our best to make this self-contained. If we refer to concepts you haven't seen in undergrad **ask us!**

Structural Engineering

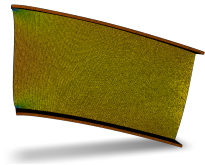
Structural engineering can be divided into three disciplines:



Analysis



Design



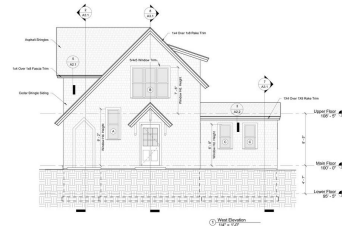
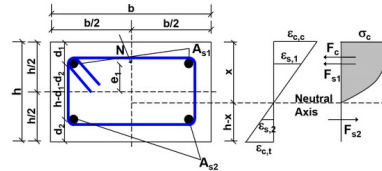
Research

These tend to be more disconnected than you would expect.

Disconnect

Disjointed because

- Practical demands often outpace mathematical understanding (*cough* concrete)
- Sometimes modeling is just not worth it.



Everything is math!

① Introduction

② Math

Algebraic Equations

Differential Equations

Uncertain Equations

③ Modeling

④ Design

⑤ Research

Algebraic Equations

Objective

State an equality in terms of an unknown variable, then solve:

- Scalar equations: find one number, say x

$$x^2 + 2x - 1 = 0$$

- Vector equations: find multiple numbers, say x, y

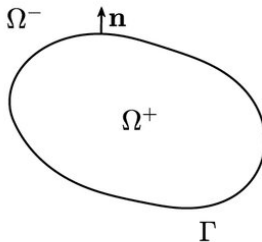
$$(1 - x)^2 + 100 (y - x^2)^2 = 0$$

Partial Differential Equations (PDE)s

Find infinitely many numbers, ie, a function $u(x, y, z, \dots)$

$$\begin{cases} \Delta u = f & \text{in } \Omega \\ u = g & \text{on } \Gamma \end{cases}$$

$$\Delta u = \nabla^2 u = \frac{d^2 u}{dx^2} + \frac{d^2 u}{dy^2} + \frac{d^2 u}{dz^2}$$



Ordinary Differential Equations (ODEs)

- Find a function $u(x)$ where x is a scalar, eg,

$$\begin{cases} u^{iv} = f(x) \\ u(0) = u_i \\ u(L) = u_j \end{cases}$$

- Boundaries of the domain $[a, b]$ are just numbers; we can use symbolic placeholders for $u(a)$ and $u(b)$, then recover by solving a *linear algebraic equation*

Linear Algebra

When the number of variables is more than two, *linear algebra* cleans things up:

Node 1: $10 + 0.866 \cdot F_{12} = 0$

$$F_{12} = \frac{-10}{0.866} = -11.547 \text{ lb}_f$$

$$0.5 \cdot F_{12} + F_{13} = 0$$

$$F_{13} = -(0.5)(-11.547) = 5.774 \text{ lb}_f$$

Node 2: $-4 - 0.866 \cdot F_{12} - 0.866 \cdot F_{23} = 0$

$$F_{23} = \frac{4 + (0.866)(-11.547)}{-0.866} = 6.928 \text{ lb}_f$$

$$-0.5 \cdot F_{12} + 0.5 \cdot F_{23} + F_{34} = 0$$

$$F_{34} = (0.5)(-11.547) - (0.5)(6.928) = -9.238 \text{ lb}_f$$

Node 3: $-4 + 0.866 \cdot F_{23} + 0.866 \cdot F_{34} = 0$

$$F_{34} = \frac{4 - (0.866)(6.928)}{0.866} = -2.309 \text{ lb}_f$$

$$-F_{13} - 0.5 \cdot F_{23} + 0.5 \cdot F_{34} + F_{35} = 0$$

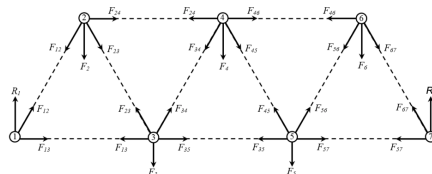
$$F_{35} = 5.774 + (0.5)(6.928) - (0.5)(-2.309) = 10.392 \text{ lb}_f$$

Node 4: $-4 - 0.866 \cdot F_{34} - 0.866 \cdot F_{45} = 0$

$$F_{45} = \frac{4 + (0.866)(-2.309)}{-0.866} = -2.309 \text{ lb}_f$$

$$-F_{34} - 0.5 \cdot F_{34} + 0.5 \cdot F_{45} + F_{46} = 0$$

$$F_{46} = -9.238 + (0.5)(-2.309) - (0.5)(-2.309) = -9.238 \text{ lb}_f$$



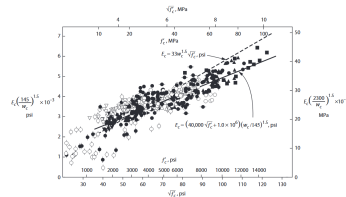
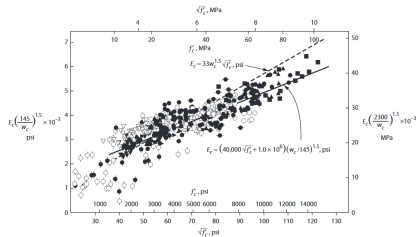
$$\begin{bmatrix} 0.5 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.866 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.5 & 0 & 0.5 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.866 & 0 & -0.866 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -0.5 & 0 & 0.5 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.866 & 0 & 0.866 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -0.5 & 0 & 0.5 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -0.866 & 0 & -0.866 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & -0.5 & 0 & 0.5 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.866 & 0 & 0.866 & 0 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \\ F_{24} \\ F_{34} \\ F_{35} \\ F_{45} \\ F_{46} \\ F_{56} \\ F_{57} \end{bmatrix} = \begin{bmatrix} 0 \\ -10 \\ 0 \\ 4 \\ 0 \\ 4 \\ 0 \\ 4 \\ 0 \\ 4 \end{bmatrix}$$

$\mathbf{B} \quad \mathbf{q} \quad \mathbf{p}$

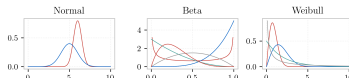
$$\mathbf{q} = \mathbf{B}^{-1} \mathbf{p}$$

Probability

- Structural engineering is plagued with uncertainty
- Probability theory formalizes this
- This can enrich modeling



Variance in material properties.



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Modeling Summary

Given loads $\mathbf{f}$, predict displacements $\mathbf{u}$.



$$\begin{cases} \operatorname{div} \boldsymbol{\sigma} + \mathbf{f} = \mathbf{0} & \text{in } \Omega \\ \mathbf{u} = \bar{\mathbf{u}} & \text{on } \Gamma_u \end{cases}$$

- CE231: PDE BVPs.
- CE222: Techniques used by software to solve PDE BVPs
- CE220: ODE, BVP
- CE225: ODE, IVP



Modeling Terminology

Conversations on modeling are often centered around three concepts:

- **Kinematics:** Aspects related to displacements and deformation; finding a way to describe *deformation* from *displacement*. *Deformation* quantifies distortion in a manner that is consistent in any coordinate system.
- **Equilibrium:** Aspects related to balance of forces, and
- **Constitutive behavior:** Characterizes material properties

All three of these are generally described independently of each other.

CE231: Continuum Mechanics

Find $\mathbf{u}(x, y, z, t)$ that satisfies

$$\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mathbf{f} = \rho \ddot{\mathbf{u}}$$

at each point x, y, z .

- Given $\mathbf{u}(x, y, z, t)$, is the equation above satisfied?
- How to choose a meaningful function σ
- Reductions to 2D

CE220: Structural Analysis

The 3D PDE BVP is approximated by a piecewise ODE:

$$\begin{aligned}\boldsymbol{\sigma}(x, y, z) &\rightarrow \mathbf{M}(x), \mathbf{N}(x) \\ \nabla_{\text{sym}} \mathbf{u} &\rightarrow \boldsymbol{\kappa}, \boldsymbol{\varepsilon},\end{aligned}$$

- In each member solve an ODE BVP with *placeholder* boundaries v_i, v_j :

$$\begin{cases} EI v^{\text{iv}} = f \\ v(0) = v_i, v(L) = v_j, \dots \end{cases} \quad (1)$$

- Use kinematics to find a *compatible* collection of boundary conditions

$$\mathbf{v} = \mathbf{B}^T \mathbf{u}$$

- Assemble a *linear algebraic equation*:

$$\boxed{\mathbf{K} \mathbf{u} = \mathbf{f} \quad \rightarrow \quad \mathbf{u} = \mathbf{K}^{-1} \mathbf{f}} \quad (2)$$

CE225: Structural Dynamics

Add acceleration to eq. (2) to recover another ODE:

Find $\mathbf{u} : \mathbb{R} \rightarrow \mathbb{R}^n$

$$\begin{cases} \mathbf{M}\ddot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f} \\ \mathbf{u}(0) = \mathbf{u}_0 \\ \dot{\mathbf{u}}(0) = \dot{\mathbf{u}}_0 \end{cases} \quad (3)$$

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Codes

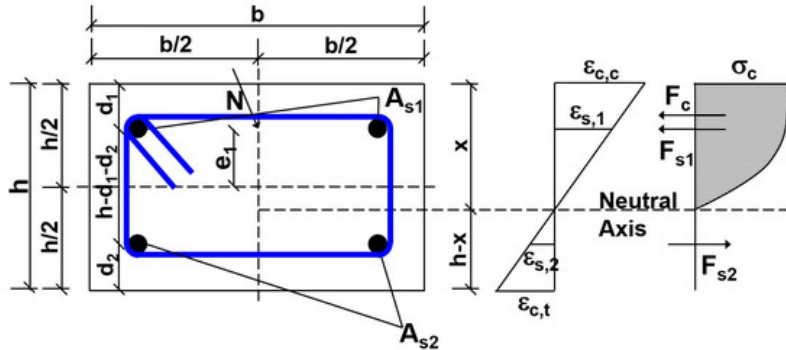
⑤ Research

Design

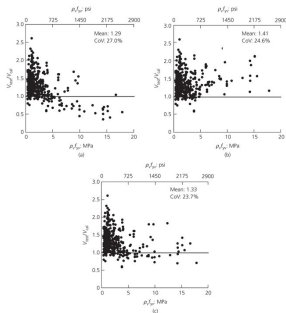
Closely related to experimentation

- CE244: Concrete
- CE247: Steel
- CE227: Earthquake resistant design

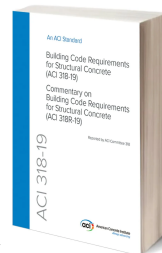
Connection to Modeling



Codes

Table 22.5.5.1— V_c for nonprestressed members

Criteria	V_c	
$A_r \geq A_{c,min}$	Either of:	$\left[2\lambda\sqrt{f'_c} + \frac{N_u}{6A_g} \right] b_w d$ (a)
		$\left[8\lambda(p_u)^{1/3} \sqrt{f'_c} + \frac{N_u}{6A_g} \right] b_w d$ (b)
$A_r < A_{c,min}$		$\left[8\lambda_3 \lambda(p_u)^{1/3} \sqrt{f'_c} + \frac{N_u}{6A_g} \right] b_w d$ (c)



ACI 318

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Welcome!